Always provide justifications and show any intermediate work for your answers. A correct but unsupported answer may not receive any marks.

Below are some equations that may be useful:

Bernoulli distribution: $$p(x|\mu) = \mu^x(1-\mu)^{1-x}, \quad x \in [0,1] \tag{1}$$

Binomial distribution: $$p(x|n,\boldsymbol{\mu}) = \binom{n}{x}\boldsymbol{\mu}^x(1-\boldsymbol{\mu})^{n-x}, \tag{2}$$

Beta distribution: $$\text{Beta}(\mu|\alpha,\beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\mu^{\alpha-1}(1-\mu)^{\beta-1} \tag{3}$$

Gamma distribution: $$\text{Gamma}(\tau|a,b) = \frac{1}{\Gamma(a)}b^a\tau^{a-1}e^{-b\tau} \tag{4}$$

Laplace distribution: $$\text{Lap}(x|\mu,b) = \frac{1}{2b}\text{exp}\left(-\frac{|x-\mu|}{b}\right) \tag{5}$$

Normal distribution: $$\mathcal{N}(x|\mu,\Sigma) = \frac{1}{(2\pi)^{D/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu)\right) \tag{6}$$

Expectation Identities

$$\mathbb{E}[AX] = A\mathbb{E}[X] = \mathbb{E}[AX] \tag{7}$$
$$\mathbb{E}[X^2] = \mathbb{E}[X]^2 + \text{Var}[X] \tag{8}$$
$$\text{Var}[X] = \mathbb{E}[XX^\top] - \mathbb{E}[X]\mathbb{E}[X]^\top \tag{9}$$
$$\text{Var}[AX] = A\text{Var}[X]A^\top \tag{10}$$
$$\text{Cov}[X,Y] = \mathbb{E}[XY^\top] - \mathbb{E}[X]\mathbb{E}[Y]^\top \tag{11}$$
$$\text{Cov}[X,Y] = \text{Cov}[Y,X]^\top \tag{12}$$
$$\text{Cov}[X,Y] = 0 \quad \text{for} \quad X \perp Y \tag{13}$$
$$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y] \tag{14}$$
$$\text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y] \quad \text{for} \quad X \perp Y \tag{15}$$
$$\mathbb{E}[XY^\top] = \mathbb{E}[X]\mathbb{E}[Y]^\top \quad \text{for} \quad X \perp Y \tag{16}$$

Inequalities and Laws:

Markov's Inequality

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a} \tag{17}$$

Chebyshev's Inequality

$$P(|X - \mathbb{E}[X]| \geq k\sigma) \leq \frac{1}{k^2} \tag{18}$$

Weak Law of Large Numbers

$$\lim_{N\to\infty} P\left(\left|\frac{1}{N}\sum_{n=1}^{N} X_n - \mu\right| < \epsilon\right) = 1 \tag{19}$$

Law of the Unconscious Statistician

$$\mathbb{E}_Y[Y] = \mathbb{E}_X[g(X)] \tag{20}$$

Gaussian Conditioning

$$p\begin{pmatrix} x \\ y \end{pmatrix} = \mathcal{N}\left(\begin{pmatrix} x \\ y \end{pmatrix}; \begin{pmatrix} m_x \\ m_y \end{pmatrix}, \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix}\right) \tag{21}$$

We have:

$$p(x|y) = \mathcal{N}\left(x; m_x + \Sigma_{xy}\Sigma_{yy}^{-1}(y - m_y), \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx}\right) \tag{22}$$

$$p(x) = \mathcal{N}(x; m_x, \Sigma_{xx}) \tag{23}$$

Woodbury Identity

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1} \tag{24}$$

Vector/Matrix Derivative Identities

$$\frac{\partial}{\partial x}a^\top x = a^\top \tag{25}$$

$$\frac{\partial}{\partial x}x^\top Bx = x^\top(B + B^\top) \tag{26}$$

$$\frac{\partial}{\partial X}\log|X| = (X^{-1})^\top = X^{-\top} \quad \text{for square } X \tag{27}$$

$$\frac{\partial}{\partial X}\log|X| = X^{-1} \quad \text{for symmetric } X \tag{28}$$

$$\frac{\partial}{\partial X}a^\top X^{-1}b = -X^{-\top}ab^\top X^{-\top} \tag{29}$$

$$\frac{\partial}{\partial \theta}X^{-1}(\theta) = -X^{-1}\frac{dX}{d\theta}X^{-1} \tag{30}$$